

Ex 1.

Write a program that reads multi-digit numbers until 0 and returns the product of the even digits of the input number

Input: Output:

12345

32

725

1237

0

```
1 n = input()
2 p = 1
3 while n != '0':  $\rightarrow n \neq 0$ 
4     for digit in n:
5         if digit in '02468': even
6             p *= int(digit) if digit is even, then p*digit
7     n = input()
8     print(p)
9
10
```

Ex 2.

Write a program that reads numbers separated by spaces until the EOF and counts the elements which are square number.

```
1 import math
2
3 while True:
4     try:
5         n = input().split()  $\rightarrow$  separate by space
6         count = 0
7         for i in n:
8             num = int(i)
9             if num >= 0 and math.sqrt(num) ** 2 == num:
10                 count += 1
11         print(count)
12
13 except EOFError:
14     break
```

2 16 5 36

2

Ex 4.

Write a program that randomly generate n elements between 15 and 66, and counts the arithmetic mean of the even values. Display it the results with two decimal

```
1 import random
2 n = int(input("n = "))
3 sum = 0
4 count = 0
5 l = []
6 for i in range(n):
7     num = random.randint(a=15, b=66)
8     if num % 2 == 0:
9         l.append(num)
10        sum += num
11        count += 1
12 print(l)
13 print("arith mean is equal to: {:.2f}".format(sum/count))
```

Ex 3

Write a program that reads string until the "END" string and deletes the space character

```
1 while True:
2     l = input().replace(_old: " ", _new: "")
3     if l == 'END':
4         break
5     print(l)
```

apple tree

1. Write a program that reads multi-digit numbers until the EOF and counts the number of '4' digits of the input number.

.../6

Input: 142341
482
524436

Output: 2
1
2

```
1 l = []
2 while True:
3     try:
4         n = input()
5         counter = 0
6         for digit in n:
7             if int(digit) == 4:
8                 counter += 1
9         l.append(counter)
10    except EOFError:
11        break
12 for r in l:
13     print(r)
```

2. Write a program that reads numbers separated by spaces until 0 and calculates the sum of the numbers, which are divisible by 7.

.../6

Input: 2 14 5 25
21
36 7 15 64
0

```
1 c = 0
2 while True:
3     try:
4         n = input("n = ").split()
5         for n in n:
6             if int(n) % 7 == 0:
7                 c += int(n)
8     except EOFError:
9         break
10 print(c)
```

3. Write a program that reads strings until the "END" string and duplicates the consonant characters. .../6

Input:	apple pear END
Output:	apppplle ppearr

```

3 while True:
4     n = input()
5     new = ''
6     if n == 'END':
7         break
8     for i in n:
9         if i not in 'aeiou':
10            new += i * 2
11        else:
12            new += i
13    print(new)

```

4. Write a program that randomly generate n elements between 25 and 85, and counts the average of the odd values. Display it the results with three decimal. .../6

Input:	n = 5 21 22 46 61 45
Output:	42.333

```

1 import random
2 n = int(input("n = "))
3 sum_odd = 0
4 count_odd = 0
5 l = []
6 for i in range(n):
7     num = random.randint(a=25, b=86)
8     if num % 2 != 0:
9         l.append(num)
10        sum_odd += num
11        count_odd += 1
12    print("odd values", l)
13    print(count_odd)
14    print("avarage: {:.3f}".format(sum_odd/count_odd))

```

Handwritten notes:

- 5 + 6 + 7
- 18 : 3
- cy. 2020
- 6

1. Write a program that reads multi-digit numbers (≥ 10) until 0 and returns the sum of the digits of the smallest input number.

Input: 12345
785
4236
0

Output: 20

```
1 n = int(input())
2 smallest = float('inf')
3
4 while True:
5     if n == 0:
6         break
7     if n < smallest:
8         smallest = n
9     n = int(input())
10
11 if smallest != float('inf'):
12     small_sum = sum(int(digit) for digit in str(smallest))
13     print(f"smallest: sum {small_sum}")
14
15
```

2. Write a program that reads three ascending numbers separated by spaces until the EOF and prints YES if the numbers are Pythagorean triplets ($a^2+b^2=c^2$) and NO otherwise.

Input: 3 4 5
2 6 8
7 24 25

Output: YES
NO
YES

```
1 n = input().split()
2 while int(n[0]) < int(n[1]) < int(n[2]):
3     try:
4         if (int(n[0])**2) + (int(n[1])**2) == (int(n[2])**2):
5             print("YES")
6         else:
7             print("NO")
8             n = input().split()
9     except EOFError:
10        break
```


	YES
3. Write a program that reads strings until the "END" string and deletes the numeric characters.	
Input:	Af1Pa2Pb4LrE Sd5Ur6CvCp8EtSS END
Output:	AfPaPbLrE SdUrCvCpEtSS

```

1 while True:
2     n = input()
3     if n == "END":
4         break
5     clean = ''.join(word for word in n if not word.isdigit())
6     print(clean)
7

```

Output:	AlfPaPbLrE SdUrCvCpEtSS
4. Write a program that randomly generate n numbers between 50 and 100 and counts the mean of the values which can be divided by 4. Display the results with two decimals. If there are no such numbers, then return 0.	
Input:	n = 5 51 72 66 91 84
Output:	78.00

```

1 import random
2 n = int(input("n = "))
3 sum = 0
4 count = 0
5 l = []
6 for i in range(n):
7     num = random.randint(a=50, b=101)
8     if num % 4 == 0:
9         l.append(num)
10        sum += num
11        count += 1
12
13 print(l)
14 print("average: {:.2f}".format(sum/count))

```

Ex 1.

Write a program that reads multi-digit numbers until 0 and returns the product of the even digits of the input number

Input: Output:

12345

32

715

1237

0

Ex 2.

Write a program that reads numbers separated by spaces until the EOF and counts the elements which are square number.

Ex 3

Write a program that reads string until the "END" string and deletes the space character

Ex 4.

Write a program that randomly generate n elements between 15 and 60, and counts the arithmetic mean of the even values. Display it the results with two decimal